

Part 1: Individual Cloud API (Required)

- **API Name** : Overall Progress API
- **Purpose** : This API displays study activity, number of summaries/quizzes/flashcards generated, score performance over time, and recent quizzes or flashcards.
- **Method (GET, POST, PUT, DELETE)** : GET
- **Cloud Service (e.g., Azure, AWS, etc.)** : Google Firestore
- **Input:**

```
{  
  "userId": "user123"  
}
```

- **Output**

```
{  
  "stats": {  
    "totalQuizzes": 12,  
    "averageScore": 75,  
    "totalSummaries": 5,  
    "totalFlashcards": 23  
  },  
  "scoreTrend": [  
    { "date": "2026-05-01", "score": 50 },  
    { "date": "2026-05-03", "score": 75 },  
    { "date": "2026-05-05", "score": 100 }  
  ],  
  "recentQuiz": {  
    "id": "quiz456",  
    "summaryTitle": "Photosynthesis",  
    "score": 2,  
    "total": 3,  
    "percentage": 67,  
    "createdAt": "2026-05-05"  
  },  
  "recentFlashcard": {  
    "id": "flash789",  
    "summaryTitle": "World History",  
    "cardCount": 8,  
    "createdAt": "2026-05-04"  
  }  
}
```

- **Related Module** : Progress Tracking Module
- **Related Database Table(s)** : Progress

Part 2: Team Cloud API

API 1: Text Summarization API

- **API Name** : Text Summarization API
- **Purpose** : This API converts user provided text or uploaded content into a concise summary to help students quickly understand key concepts
- **Method (GET, POST, PUT, DELETE)** : POST
- **Cloud Service (e.g., Azure, AWS, etc.)** : Deployed on Vercel
- **Input**

Format: multipart/form-data

Key	Type	Description
text	String	Text to summarize
file	File (optional)	Uploaded document
mode	String	Operation type (e.g., "summary")

- **Output**

```
{  
  "result": "Artificial Intelligence helps automate tasks and improve efficiency..."  
}
```

- Related Module : Summarization AI Agent
- Related Database Table(s) : Content & Summary

API 2: Video Summarization API

- **API Name** : Video Summarization API
- **Purpose** : This API extracts and summarizes key information from video content (such as lectures or YouTube videos) to help students review material efficiently.
- **Method (GET, POST, PUT, DELETE)** : POST
- **Cloud Service (e.g., Azure, AWS, etc.)** : Deployed on Vercel
- **Input**

Format: JSON

```
{  
  "url": "https://www.youtube.com/watch?v=example"  
}
```

- **Output**

```
{  
  "summary": "This video explains key concepts of Artificial Intelligence including  
automation and efficiency..."  
}
```

- **Related Module** : Video/Lecture Summarization Agent
- **Related Database Table(s)** : Content, Summary & Video

Part 3: Request & Response

API 1: Text Summarization API

Endpoint:

<https://summarize-lemon.vercel.app/api/summarize>

Method:

POST

Request Format:

This API uses form - data (NOT JSON)

Body

Key	Value
text	Any paragraph you want to summarize
mode	summary

Example Input

text :

Artificial Intelligence helps automate tasks and improve efficiency.

mode :

summary

Expected Output :

```
{  
  "result": "Artificial Intelligence helps automate tasks and improve efficiency by improving productivity and reducing manual work."  
}
```

Error Case 1: Missing Text

```
{  
  "error": "Text input is required"  
}
```

Error Case 2: Wrong Format (JSON instead of form-data)

```
{  
  "error": "Invalid request format"  
}
```

API 2: Video Summarization API

Endpoint:

<https://summarize-lemon.vercel.app/api/video-summarize>

Method:

POST

Request Format:

JSON

Request Body

```
{  
  "url": "https://www.youtube.com/watch?v=example"  
}
```

Example Input

url: https://www.youtube.com/watch?v=example

Expected Output :

```
{  
  
  "summary": "This video explains how Artificial Intelligence works and how it improves efficiency  
through automation and data processing."  
}
```

Error Case 1: Missing URL

```
{  
  "error": "Video URL is required"  
}
```

Error Case 2: Invalid or Unsupported Video

```
{  
  "error": "Unable to extract transcript from video"  
}
```

Part 4: Peer API Testing (In-Class Activity)

Each team will test another team's APIs AS WE LINED UP LAST WEEK

During the session:

- Review assigned team's APIs
- Check clarity of input/output
- Validate logic and correctness
- Test edge/error cases

Part 5: Testing Results

Team Tested: Game Start API Team **APIs Tested:** Games List API — GET
<https://game-start-api-v1.onrender.com/api/games>

Games List API

What is good

- Endpoint is live, returns valid JSON, no auth needed — easy to test.
- Response includes solid game data: name, rating, platforms, genres, screenshots, etc.
- Includes pagination ([count](#), [next](#), [previous](#)) and uses proper GET method.

Issues found

- **Exposed API key.** The [next](#) URL leaks their RAWG API key in plaintext — needs to be fixed ASAP.
- **It's just a RAWG proxy.** Response is passed through unchanged with no real value-add layer.
- **No documentation.** No input fields, error cases, or sample requests provided.
- **Response is bloated.** Returns full system requirements and HTML for every game in the list.
- **No error cases testable.** Only the root endpoint was shared.

Suggestions for improvement

1. Hide the RAWG API key server-side and rewrite pagination URLs to their own domain.
2. Add query params for filtering/searching ([?search=](#), [?genre=](#), [?platform=](#)).
3. Add a detail endpoint ([GET /api/games/:id](#)) so there's more than one route.
4. Trim the list response — move heavy data to the detail endpoint.
5. Document the API properly with sample requests, responses, and error cases.